

WEEK-4

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Servers' tree is expanded to show 'PostgreSQL 18' > 'Databases (2)' > 'bookflow_db' > 'postgres'. The main panel shows a SQL query being executed in the 'Query' tab. The query is as follows:

```
1 CREATE EXTENSION IF NOT EXISTS vector;
2 SELECT extname, extversion
3 FROM pg_extension
4 WHERE extname = 'vector';
5 CREATE TABLE books (
6     id SERIAL PRIMARY KEY,
7     title VARCHAR(200) NOT NULL,
8     author VARCHAR(200),
9     description TEXT
10 );
11
```

The 'Messages' tab at the bottom shows the execution result: 'CREATE TABLE' and 'Query returned successfully in 78 msec.' The status bar at the bottom indicates 'Total rows: 1', 'Query complete 00:00:00.078', 'CRLF', and 'Ln 11, Col 1'.

Object Explorer | Servers (1) | PostgreSQL 18 | Databases (2) | bookflow_db | postgres | Casts | Catalogs | Event Triggers | Extensions | Foreign Data Wrappers | Languages | Publications | Schemas | Subscriptions | Login/Group Roles | Tablespaces | pgAgent Jobs

Query: bookflow_db/postgres@PostgreSQL 18

```

11 WHERE extname = 'vector';
12 CREATE TABLE books (
13   id SERIAL PRIMARY KEY,
14   title VARCHAR(200) NOT NULL,
15   author VARCHAR(200),
16   description TEXT
17 );
18 INSERT INTO books (title, author, description) VALUES
19 ('The Martian', 'Andy Weir', 'A stranded astronaut struggles to survive on Mars.'),
20 ('Dune', 'Frank Herbert', 'A science-fiction story involving planets, politics and survival.'),
21 ('Pride and Prejudice', 'Jane Austen', 'A classic story of relationships, family and society.'),
22 ('The Hobbit', 'J.R.R. Tolkien', 'A fantasy adventure involving a journey, creatures and a dang
23 ('Foundation', 'Isaac Asimov', 'A science-fiction story about humanity and the future of civili
24 SELECT * FROM books;

```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

id	title	author	description
1	The Martian	Andy Weir	A stranded astronaut struggles to survive on Mars.
2	Dune	Frank Herbert	A science-fiction story involving planets, politics and survival.
3	Pride and Prejudice	Jane Austen	A classic story of relationships, family and society.
4	The Hobbit	J.R.R. Tolkien	A fantasy adventure involving a journey, creatures and a dangerous qu...
5	Foundation	Isaac Asimov	A science-fiction story about humanity and the future of civilization.

Object Explorer | Servers (1) | PostgreSQL 18 | Databases (2) | bookflow_db | postgres | Casts | Catalogs | Event Triggers | Extensions | Foreign Data Wrappers | Languages | Publications | Schemas | Subscriptions | Login/Group Roles | Tablespaces | pgAgent Jobs

Query: bookflow_db/postgres@PostgreSQL 18

```

14 ('Pride and Prejudice', 'Jane Austen', 'A classic story of relationships, family and society.')
15 ('The Hobbit', 'J.R.R. Tolkien', 'A fantasy adventure involving a journey, creatures and a dang
16 ('Foundation', 'Isaac Asimov', 'A science-fiction story about humanity and the future of civili
17 SELECT * FROM books;
18 ALTER TABLE books ADD COLUMN embedding VECTOR(3);
19 UPDATE books
20 SET embedding = (CASE title
21   WHEN 'The Martian' THEN '[0.10, 0.90, 0.20]'
22   WHEN 'Dune' THEN '[0.20, 0.85, 0.25]'
23   WHEN 'Pride and Prejudice' THEN '[0.80, 0.10, 0.70]'
24   WHEN 'The Hobbit' THEN '[0.30, 0.20, 0.90]'
25   WHEN 'Foundation' THEN '[0.15, 0.80, 0.30]'
26 END)::vector;
27 SELECT id, title, embedding FROM books;

```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

id	title	embedding
1	The Martian	[0.10,0.90,0.20]
2	Dune	[0.20,0.85,0.25]
3	Pride and Prejudice	[0.80,0.10,0.70]
4	The Hobbit	[0.30,0.20,0.90]
5	Foundation	[0.15,0.80,0.30]

Successfully run. Total query runtime: 119 msec. 5 rows affected.

Total rows: 5 | Query complete 00:00:00.119

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Query: bookflow_db/postgres@PostgreSQL 18

```

30 title,
31 embedding <-> '[0.10, 0.85, 0.20]' AS l2_distance
32 FROM books
33 WHERE embedding IS NOT NULL
34 ORDER BY embedding <-> '[0.10, 0.85, 0.20]'
35 LIMIT 3;
36 SELECT
37   id,
38   title,
39   embedding <=> '[0.10, 0.85, 0.20]' AS cosine_distance
40 FROM books
41 WHERE embedding IS NOT NULL
42 ORDER BY embedding <=> '[0.10, 0.85, 0.20]'
43 LIMIT 3;

```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

id	title	cosine_distance
1	The Martian	9.408262185794936e-05
2	Dune	0.0072885302161147525
3	Foundation	0.009750970439461937

Successfully run. Total query runtime: 146 msec. 3 rows affected.

Total rows: 3 | Query complete 00:00:00.146

```
1 openapi: 3.0.3
2 info:
3   title: BookFlow Search API
4   version: 1.0.0
5   description: API contract for semantic book search
6 paths:
7   /v1/books/search:
8     get:
9       summary: Search books semantically
10      operationId: searchBooks
11      parameters:
12        - name: q
13          in: query
14          required: true
15          description: Natural-language search query
16          schema:
17            type: string
18            minLength: 1
19        - name: limit
20          in: query
21          required: false
22          description: Maximum number of results
23          schema:
24            type: integer
25            minimum: 1
26            maximum: 20
27            default: 3
28      responses:
29        '200':
30          description: Search results
31          content:
32            application/json:
33              schema:
34                type: object
35                required: [query, results]
36                properties:
37                  query:
38                    type: string
39                  results:
40                    type: array
41                    items:
42                      type: object
43                      required: [id, title, score]
44                      properties:
```

default

GET /v1/books/search Search books semantically

Try it out

Name	Description
q * required string (query)	Natural-language search query
limit integer (query)	Maximum number of results Default value : 3

Responses

Code	Description	Links
200	Search results	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "query": "string",
  "results": [
    {
      "id": 0,
      "title": "string",
      "score": 0
    }
  ]
}
```

Swagger Editor

```
1 openapi: 3.0.3
2 info:
3   title: BookFlow Search API
4   version: 1.0.0
5   description: API contract for semantic book search
6 paths:
7   /v1/books/search:
8     get:
9       summary: Search books semantically
10      operationId: searchBooks
11      parameters:
12        - name: q
13          in: query
14          required: true
15          description: Natural-language search query
16          schema:
17            type: string
18            minLength: 1
19        - name: limit
20          in: query
21          required: false
22          description: Maximum number of results
23          schema:
24            type: integer
25            minimum: 1
26            maximum: 20
27            default: 3
28      responses:
29        '200':
30          description: Search results
31          content:
32            application/json:
33              schema:
34                type: object
```

BookFlow Search API 1.0.0 OAS 3.0

API contract for semantic book search

default

GET /v1/books/search Search books semantically

Try it out

Name	Description
q * required string (query)	Natural-language search query
limit integer (query)	Maximum number of results Default value : 3

Responses

Code	Description	Links
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Example Value | Schema

```
{
  "query": "string",
  "results": [
    {
      "id": 0,
      "title": "string",
      "score": 0
    }
  ]
}
```

Swagger Editor

```
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29        '200':
30          description: Search results
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33              schema:
34                type: object
```

Responses

Code	Description	Links
200	Search results	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "query": "string",
  "results": [
    {
      "id": 0,
      "title": "string",
      "score": 0
    }
  ]
}
```

400	Bad Request	No links
-----	-------------	----------

Media type: application/json

Example Value | Schema

```
{
  "error": {
    "code": "string",
    "message": "string"
  }
}
```

Swagger Editor